

Xiling: Cryo-CMOS Manipulator Using Dual 18-bit R-2R DACs for Single-Electron Transistor at 60 mK

Yingjie Li^{ID}, Yifei Zhang^{ID}, Haichuan Lin^{ID}, and Cheng Wang^{ID}, *Senior Member, IEEE*

Abstract—Millions of quantum-bits (qubits) are envisioned for a fault-tolerant quantum computer. For scalability, silicon spin qubit stands out due to its compatibility with advanced CMOS processes. Silicon single-electron transistors (SETs) are widely adopted for the quantum state discrimination of spin qubits. For high-fidelity quantum logic gate and readout, the gate electrodes of the above Si quantum devices require a biasing voltage with $\sim 10\text{-}\mu\text{V}$ precision to define the electron occupation, and ultra-low noise to suppress the charge noise induced decoherence. In addition, the co-integration of a large-scale spin qubit array and cryogenic CMOS (Cryo-CMOS) electronics on a $<100\text{-mK}$ stage of dilution refrigerator requires ultra-low power (LP). This article presents a 65-nm Cryo-CMOS manipulator integrated with a SET at 60 mK, consuming a DC power of 60 μW only. It adopts a pair of 18-bit R-2R DACs for the gate control. Each DAC has 63 bits of MSB thermal code, 3 bits of MID thermal code, 10 bits of binary code, and 3 bits of fine-tuning code. However, at cryogenic temperature, the P+ polysilicon resistors of the DAC suffer from $43\times$ stronger nonlinearity and $2\times \sim 3\times$ worse device mismatch of the resistance value, limiting integral nonlinearity (INL) and differential nonlinearity (DNL). Accordingly, resistor nonlinearity calibration (RNC) is proposed to compensate for the voltage dependency of resistance using a pair of tailed pMOS and nMOS transistors. Ordered element matching (OEM) is also adopted to mitigate random mismatch. Measured at 4 K, this chip achieves a state-of-the-art 18-bit resolution with $4.6\text{-}\mu\text{V}$ precision, an INL of ± 0.8 LSB, and a DNL of ± 0.8 LSB. It also exhibits an integrated noise of $3.8\text{ }\mu\text{V}$ and a noise spectrum density of $4.1\text{ nV/Hz}^{0.5}$ at 60 mK. Finally, while the two DACs of this chip control the gate and source of the SET, respectively, a clear Coulomb diamond of the SET is measured at 60 mK.

Index Terms—Calibration, cryogenic CMOS (Cryo-CMOS), quantum computing, R-2R DAC, single-electron transistor (SET).

I. INTRODUCTION

QUANTUM computing potentially will achieve exponential acceleration of specific algorithmic problems that remain computationally infeasible for classical computers [1], [2]. However, a fault-tolerant quantum computer needs

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Yingjie Li and Cheng Wang are with the School of Electronic Science and Engineering (ESE), University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: wangch87@uestc.edu.cn).

Yifei Zhang is with the Southern University of Science and Technology, Shenzhen 518055, China.

Haichuan Lin is with Chengdu Data Automation System Technologies, Chengdu 610213, China.

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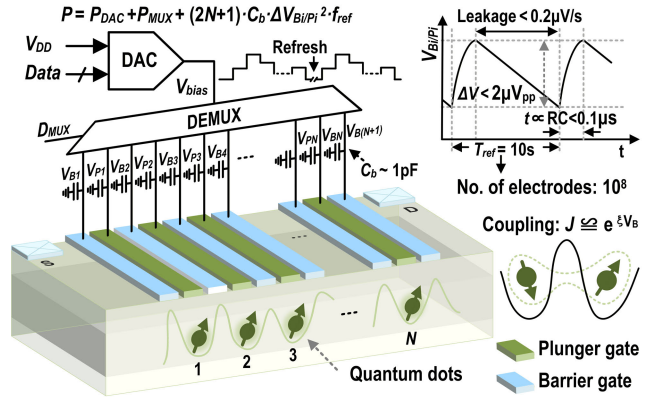


Fig. 1. Multiplexed gate biasing for massive Si qubit array.

$10^6 \sim 10^9$ physical quantum-bits (qubits) for quantum error correction (QEC) [2]. Thanks to the compatibility with the advanced silicon process, silicon spin qubits are scaling-friendly, compared with superconducting or trapped-ion qubits. Currently, most Si qubits are hosted on the $<100\text{-mK}$ stage of dilution refrigerators for better fidelity, which only has $100\text{ }\mu\text{W} \sim 1\text{-mW}$ thermal handling capacity [3]. Future co-integration of a large-scale spin qubit array with ultra-low power (LP) cryogenic CMOS (Cryo-CMOS) control electronics holds the potential of wafer-scale quantum processor units (QPUs) working above 1 K [4]. Intel has already revealed the fabrication and 1.6-K verification of hundreds of Si/SiGe quantum dots (QDs) on a 300-mm Si wafer [5].

As shown in Fig. 1, each Si QD, built either by a single phosphorus atom [6] or a Si/SiGe [7] hetero-structure, needs at least two electrostatic biasing gates: plunger gate and barrier gate. Although previous works indicate a precision level of $100\text{ }\mu\text{V}$ can be sufficient to define the chemical potential and regulate the coupling of Si qubits [8], a higher precision of $10\text{ }\mu\text{V}$ can provide additional benefits, which compensates the source-drain offset of quantum device, and the voltage loss due to the charge leakage on the gate capacitor [4]. In addition, the voltage noise induces energy-level fluctuation, reducing the coherence and entanglement fidelity of QDs [9]. Furthermore, conventional room-temperature instruments at 300-K encounter severe scalability limits due to sophisticated cabling, crosstalk, and heat leakage [1]. Therefore, monolithic integration [10] or co-packaging [11] of Si qubits and cryo-CMOS control electronics becomes promising. Since operation at $<100\text{ mK}$ requires ultra-LP, time-domain multiplexing for

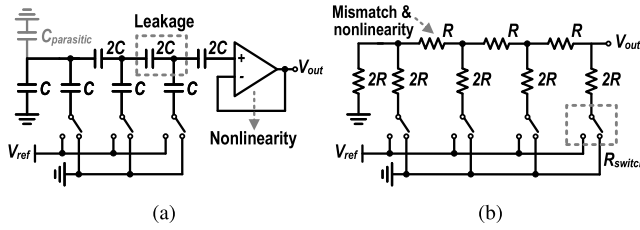


Fig. 2. Schematic of (a) CDAC and (b) resistive R-2R DAC.

gate biasing has been proposed (Fig. 1) [12], [13], [14], [15], [16], [17]. Considering the reduced leakage of capacitors ($< 0.2 \mu\text{V/s}$) [14] at cryogenic temperature, the voltage drift (ΔV) on the capacitor is less than $2 \mu\text{V}$ with a refresh rate f_{ref} of 0.1 Hz. During this refresh period, a DAC can theoretically bias up to 10^8 electrodes (5×10^7 QDs) at a demultiplexing rate of 10 MHz with the sample-and-hold capacitor C_b of 1 pF. In addition, due to the low-voltage drift and low refresh rate, the power consumption of each electrode capacitor is extremely low ($P = C_b \cdot \Delta V^2 \cdot f_{ref}$).

For high precision voltage bias, Cryo-CMOS capacitive DAC (CDAC) (CDAC) have been adopted [8], [10], [12], [13], [18], [19]. Co-integration of CDACs with the Si QD or single-electron transistors (SETs) has been demonstrated at 44 mK [13] and 4.2 K [10]. However, although enjoying a compact chip area, the CDACs exhibit limited precision due to parasitic capacitance and charge leakage [Fig. 2(a)]. In most cases, buffer op-amps are also necessary to drive the load, introducing extra nonlinearity, DC power overhead, and speed limit [19]. An exception is the op-amp in [8] and [18], which is turned off during the qubit biasing and only enabled during the DAC performance measurement, at the expense of longer conversion time. In addition, the output noise of the buffer op-amps is undermined by shot noise [20] and flicker noise [21] of cryogenic MOSFET. Therefore, a 15-bit Cryo-CMOS CDAC at 4.2 K presents an integral nonlinearity (INL) of 36.5 LSB, a differential nonlinearity (DNL) of 0.8 LSB and an integrated noise of $188 \mu\text{V}$ [12]. DNL of the 13-bit [13] and 11-bit [19] Cryo-CMOS CDACs for quantum devices only reach 2.5 LSB and 2 LSB, respectively. Although a DNL of < 1 LSB can ensure the monotonically increasing of the DAC biasing voltage for the Si qubit, an INL of < 1 LSB achieves repeatable biasing of the same qubit with different DACs. This is critical for a large-scale qubit array. A good INL also benefits the time-domain leakage compensation with the uniform minimum voltage steps. Furthermore, classic current steering DACs are suitable for high-speed applications [22]. But they present relatively low precision, high DC power, and severe deterioration of flicker noise at cryogenic temperatures. For instance, an 11-bit current steering DAC in [23] exhibits an output integrated noise of $200 \mu\text{V}_{rms}$ at 4 K. Alternatively, resistive R-2R ladder DACs present higher precision and lower output noise [Fig. 2(b)]. Its output noise is mainly thermal noise of resistors, linearly decreasing with temperature. At 300 K, ADI's 20-bit R-2R DAC achieves a $20\text{-}\mu\text{V}$ precision within a $\pm 10\text{-V}$ output swing and a $1.7\text{-}\mu\text{V}$ integrated noise by employing force and sense calibration of switch resistance [24], [25]. But high DC power of 84 mW hinders

its application for spin qubits. The Cryo-CMOS R-2R DACs at 77 K [26], [27], and 4 K [17] were designed for trapped-ion qubits, presenting 12- and 11-bit resolution, respectively. More importantly, after cooled down to 4 K, the on-chip resistors present not only a resistance change up to $\pm 20\%$ [28], but also a strong voltage nonlinearity due to self-heating [29], [30], and a deteriorated random mismatch [31]. Therefore, Cryo-CMOS DACs with $10\text{-}\mu\text{V}$ -level precision have not been achieved yet.

In this work, a 65-nm Cryo-CMOS SET manipulator with a pair of 18-bit R-2R DACs is reported [32]. Employing a segmented R-2R ladder, resistor nonlinearity calibration (RNC) and ordered element matching (OEM), it exhibits a state-of-the-art 18-bit resolution with $4.6\text{-}\mu\text{V}$ precision at 4 K. It has also achieved so far the lowest integrated noise of $3.8 \mu\text{V}$ and noise spectrum density of $4.1 \text{ nV/Hz}^{0.5}$ at 60 mK. It consumes a DC power of $60 \mu\text{W}$ only. Finally, after co-integrated with SET at 60 mK, a clear Coulomb diamond is measured. This article is organized as follows. Section II elaborates the operation principle of SET. Section III describes the architecture of the dual-DAC SET manipulator. The DAC and its calibration are elaborated in Section IV. Section V presents the measured electrical performance and SET experiment. Finally, Section VI summarizes this article.

II. OPERATION PRINCIPLE OF SILICON SET AND SEP

A Si SET is a quantum device that exploits the mechanism of Coulomb blockade to control the transportation of a single electron. It plays an important role in the Elzerman readout of Si Qubit states as an ultra-sensitive charge detector [33]. As shown in Fig. 3(a), it consists of a metal island as a QD, flanked by two tunnel junctions connecting the source and drain, respectively, and a gate to control the electrochemical potential of the QD [34]. Its operation principle can be described as follows. First, in a special case where the Fermi levels of the source and drain of the SET are equal, only if the chemical potential of the island aligns with the source and drain, a single electron can be sent from the source to the drain individually. By sweeping the gate voltage V_g , which controls the chemical potential of the island, a periodic drain current I_{sd} can be observed due to the discreteness of energy levels of the island. This is called Coulomb oscillation [Fig. 3(b)]. Second, in a more general case, if the Fermi levels of source and drain are unequal, single-electron transportation will occur when V_g lies in the source-drain bias window, resulting in a Coulomb diamond [Fig. 3(c)]. Furthermore, during the Elzerman readout process, the island of SET is capacitively coupled to the QD nearby. The electrochemical potential of the island should be set between the spin-up and spin-down states of the electron on the QD [Fig. 3(d)]. Thus, only the electron in the spin-up state can perform tunneling, generating the detectable current pulse. The precision of V_g is determined by the Zeeman energy splitting E_Z between spin-up and spin-down states ($E_Z = g \cdot \mu_B \cdot B$, $g \approx 2$ is the spin gyromagnetic ratio, μ_B is the Bohr magneton) [35]. When the magnetic induction density (B) ranges from 1 to 3 T, the energy splitting is $100 \sim 350 \mu\text{eV}$. Hence, the voltage step of V_g (ΔV_g) needs to be less than 1 mV with the level arm $\alpha = 0.1$ ($\Delta V_g = E_Z/(\alpha \cdot e)$) [36]. The required voltage precision should be set as $100 \mu\text{V}$, which is one-tenth

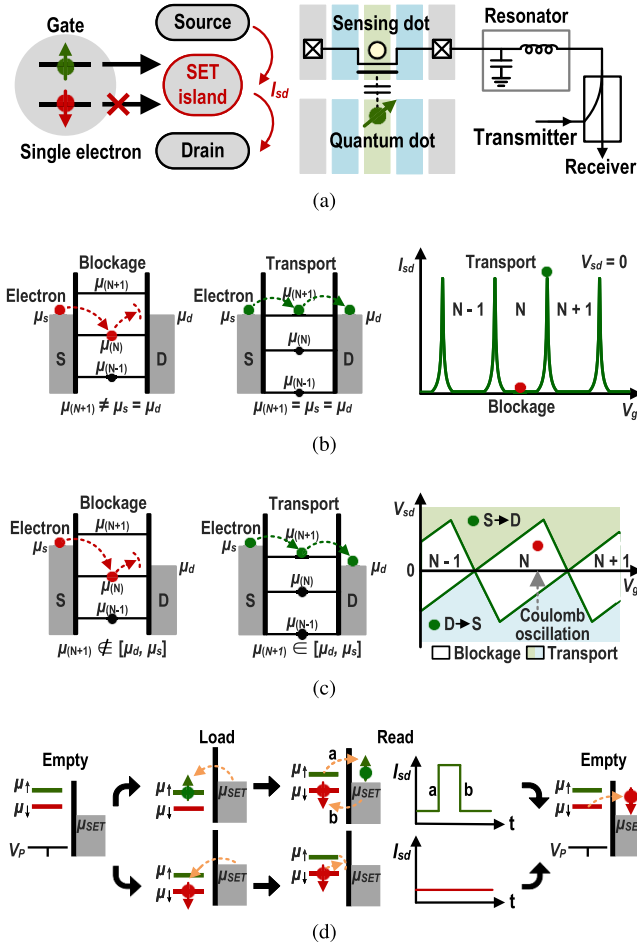


Fig. 3. (a) Si SET for Elzerman readout. (b) Coulomb oscillation. (c) Coulomb oscillation diamond. (d) Principle of Elzerman readout.

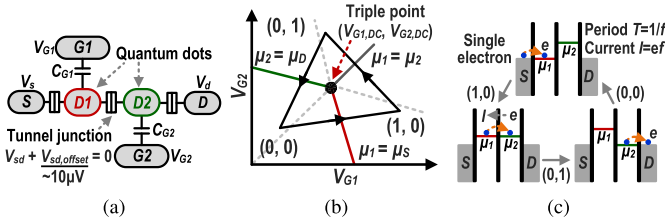


Fig. 4. (a) Block diagram of SEP. (b) Gate control pulses. (c) Periodic transportation of a single electron.

of ΔV_g and is acceptable. This precision is also suitable for the source-drain bias, sufficient to bias the SET in a state where the Coulomb oscillation current peaks exist.

In addition, the single-electron pump (SEP) is also an important Si device, which incorporates two QDs (Fig. 4). It transports a single electron under a fixed period $T = 1/f$, which is well suited for a high precision current source $I = e \cdot f$ [37]. Here, e is the electron charge. Its operation necessitates the precise biasing of two gates to set the static operating point near the triple point. Subsequently, two-phased AC pulses are applied to move the operating point periodically around the triple point, transporting one electron per cycle. Furthermore, a source-drain bias with 10- μ V precision is required to counteract the undesired source-drain offset from the measurement system and test the stable charge states [38]. In addition, the

ultra-low voltage noise is required to avoid introducing the charge noise that could flood the small output current of SEP (approximately 1.6 pA with $f = 10$ MHz). Driving the SEP using cryogenic DACs with ultra-LP consumption will reduce the thermal leakage and noise introduced by cables connecting to room-temperature instruments, allowing the purer output current of SEP and improving long-term stability. Collectively, the gate pulsing with high-precision, ultra-low noise, and ultra-LP consumption is crucial for the Si quantum devices.

III. ARCHITECTURE OF DUAL-DAC SET MANIPULATOR

In this work, a Cryo-CMOS manipulator with a pair of 18-bit R-2R DACs co-integrated with the Si SET at 60 mK is proposed. It can also serve as a SEP and a spin qubit. The chip architecture is shown in Fig. 5. Two calibrated 18-bit R-2R DACs control the source and gate of a SET, respectively. The drain of the SET is fed to the off-chip transimpedance amplifier (TIA) at 300 K for the current readout. On the analog part, the resistive R-2R ladder of each 18-bit DAC consists of 63-bit MSB thermal code $H<62:0>$, 3-bit MID thermal code $M<2:0>$, 10-bit binary code $B<9:0>$, and 3-bit LSB calibration code $L<2:0>$. This segmentation structure is motivated by a tradeoff between area and precision. The unary thermal code offers higher precision but exhibits exponential area growth with the number of bits. The binary code saves area but sacrifices precision. Therefore, thermal code is employed for the more critical high-weighted bits, segmented into 3-bit MID and 6-bit MSB to conserve area. The remaining 10-bit employs binary code, which typically requires no calibration when providing a resolution below 10-bit. Finally, an additional 3-bit LSB is added for the fine calibration.

To ensure high precision, first, a pair of pMOS and nMOS transistors is connected to each thermal code resistor leg, whose resistance is controlled by two low-precision 10-bit R-2R auxiliary DACs for the proposed RNC. RNC is then performed over the thermal code within 64 equally divided regions of the DAC's full output range ($0 \text{ V} \sim V_{DD} = 1.2 \text{ V}$). The residual nonlinearity of binary code $B<9:0>$ is calibrated by the fine calibration code $L<2:0>$. Second, to mitigate the device mismatch, instead of random element matching [dynamic element matching (DEM)] [39], OEM [40] is employed to avoid voltage disturbance induced by randomization. A specific OEM calibration procedure has been carefully developed.

The digital part of this chip has three built-in operation modes: 1) DC biasing mode, which delivers a calibrated static output voltage for the DC biasing of quantum devices; 2) two-point modulation mode for the cryogenic calibration of the 18-bit DAC, which periodically toggles between two specific DAC codes to eliminate the lower frequency noise and interference of the refrigerator through lock-in detection; 3) three-point modulation mode for a SEP device: each of the dual-DAC could periodically rotate along three specific voltages. Two DACs under three-point modulation mode could perform the adiabatic transportation of a single electron of a SEP device. Then, the DAC outputs are filtered by a

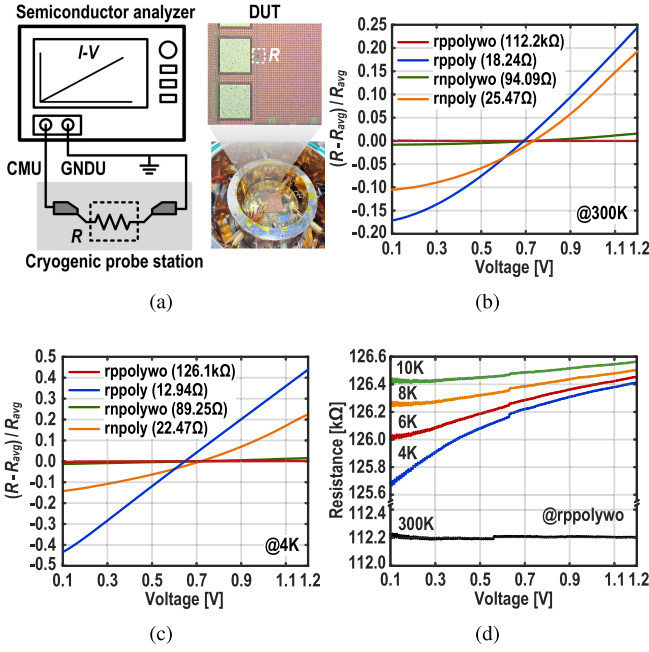


Fig. 7. (a) Measurement setup of resistor nonlinearity. Measured voltage nonlinearity of resistors at (b) 300 K and (c) 4 K. (d) Measured voltage nonlinearity of rppolywo at different temperatures.

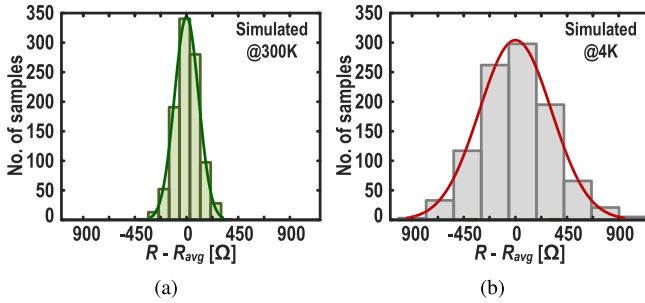


Fig. 8. Simulated R mismatch at (a) 300 K and (b) 4 K.

employed for the DAC. Fig. 7(d) indicates that the measured average resistance of R is 112.2 kΩ at 300 K with a voltage nonlinearity of 15.5 Ω/V. In contrast, the average resistance rises to 126.1 kΩ (12.4% higher) at 4 K with a voltage nonlinearity of 664 Ω/V (approximately 43× worse). It is also observed that resistor nonlinearity varies significantly from 4 to 10 K. To reduce DC power and enhance device matching, four identical polysilicon resistors are cascaded as a single resistance unit $4R$ (Fig. 5). DAC's total output impedance is then 14.0 kΩ at 300 K and 15.8 kΩ at 4 K, which leads to a time constant of $\tau = 14 \sim 16$ ns with a 1-pF load capacitor (Fig. 1).

2) *Resistor Mismatch*: The carrier mobility of the resistor is affected by various scattering mechanisms: ionized impurity scattering, grain boundary scattering, and lattice thermal vibrations. At cryogenic temperature, the uniform thermal vibrations of the lattice are almost frozen, while the remaining scattering mechanisms become sensitive to microscopic variations inherent in the resistor manufacturing process [31]. In this way, process-induced differences between

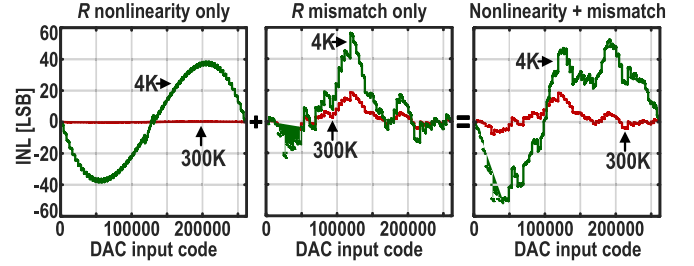


Fig. 9. Simulated INL of 18-bit DAC due to R non-ideality.

adjacent resistors are amplified, worsening the random resistor mismatch. To be specific, for P-diffusion resistors, partially doped atoms remain un-ionized, forming electrically neutral scattering centers. The number of these neutral impurities is extremely sensitive to the local doping fluctuations. For polysilicon resistors, since the thermal kinetic energy of carriers is insufficient to overcome the potential barrier of the grain boundary, they can only cross the barrier through the field-induced tunneling effect. As the potential barrier fluctuates with the grain size at cryogenic temperature, the process-induced mismatch is increased. The direct experimental quantification of the mismatch is challenging due to the impractical sample size requirement. Based on the prior work [31], it can be speculated that the resistor mismatch might deteriorate by a factor of $2 \sim 3$ at cryogenic temperatures. It was later verified in the measurement that this assumption is reasonable. The Monte-Carlo simulation of the resistor R at 300 K is performed, and the standard deviation of its mismatch is found to be 134 Ω [Fig. 8(a)]. Leveraging the temperature-dependent mismatch scaling law, the cryogenic behavior model for R is subsequently constructed. In this model, the key mismatch parameter, namely, the standard deviation of the resistance mismatch distribution, is calibrated to three times the value observed at 300 K [Fig. 8(b)].

Simulation of the DAC's linearity is conducted according to the above resistor non-ideality. It can be seen from Fig. 9 that nonlinearity itself leads to a ± 40 LSB INL on an 18-bit resolution. The mismatch itself leads to $2 \times \sim 3 \times$ worse INL. The combined effect significantly degrades the DAC linearity. Therefore, effective calibration methods are required. Due to the thermal and binary code segmentation, the DAC is not intrinsically monotonic. Thus, analog and digital calibration technologies are adopted simultaneously to reduce the storage for calibration codes and power consumption.

B. Resistor Nonlinearity Calibration

Fig. 7(d) indicates that the measured resistance of R varies by 200 ~ 800 Ω within 0 ~ 1.2 V from 10 to 4 K. Since the nonlinear variation is approximately proportional to the voltage, the thermal code resistor $8R$ employed in the DAC also varies by 200 ~ 800 Ω within 0 ~ 1.2 V. To compensate for the associated voltage dependency, a pair of a pMOS transistor M1 and an nMOS transistor M2 is introduced to each of the 63-bit MSB $H<62:0>$ thermal code resistors, as shown in Fig. 10. Two binary R-2R ladder auxiliary DACs with 10-bit resolution enable precise control of the ON-resistance of M1

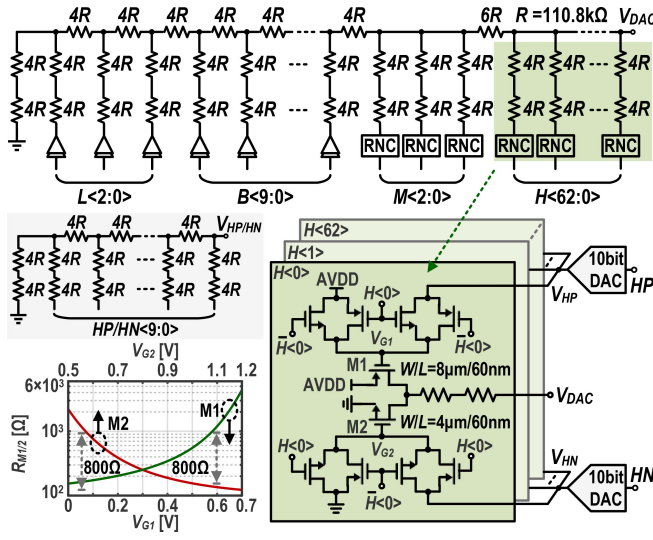


Fig. 10. Proposed RNC technology.

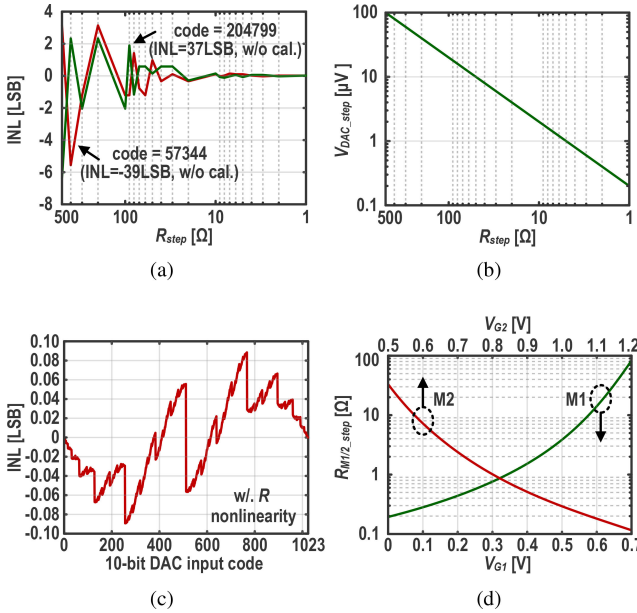


Fig. 11. Simulation results. (a) Minimum resistance step R_{step} to be adjusted during RNC versus the worst simulated INL. (b) Output voltage step V_{DAC_step} of DAC under the given R_{step} . (c) INL of 10-bit auxiliary DAC w/ R nonlinearity at 4 K. (d) Adjustable resistance step $R_{M1/2_step}$ under the gate voltages V_{G1} and V_{G2} .

and M2, respectively. When one of the 63 MSB thermal code resistors is supposed to be connected to “1,” the gate of M1 is biased to V_{HP} by the auxiliary DAC, and M2 is shut down by being directly biased to ground. Similarly, when the thermal code resistor is supposed to be connected to “0,” the gate of M2 is biased to V_{HN} from another auxiliary DAC, and M1 is shut down. The two auxiliary DACs adjust the ON-resistance of M1 and M2 to vary inversely to the voltage dependency of the polysilicon resistors, achieving constant resistance over the full output range of V_{DAC} . In addition, similar pMOS M3 and nMOS M4 transistor pairs are added to the 3-bit MID thermal code $M<2:0>$, which are governed by another 2 10-bit DACs (Fig. 5).

The size of pMOS M1 ($W/L = 8 \mu\text{m}/60 \text{ nm}$) and nMOS M2 ($W/L = 4 \mu\text{m}/60 \text{ nm}$) is carefully designed. Working in

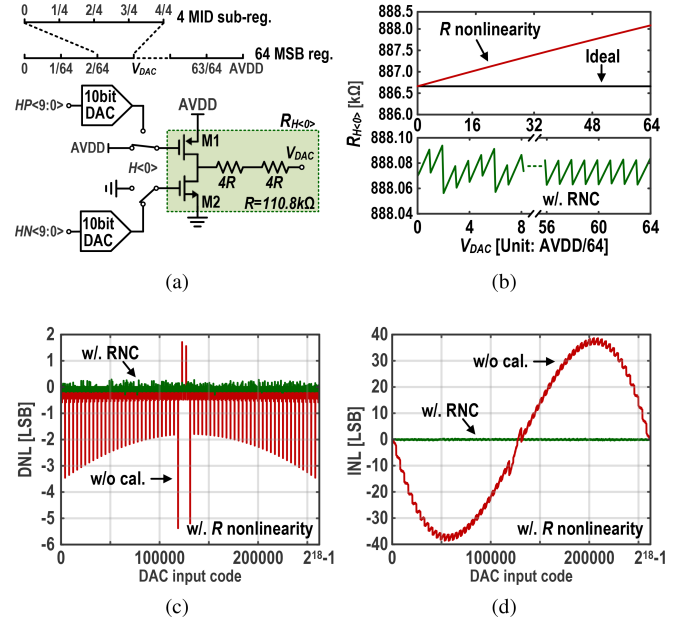


Fig. 12. (a) Segmentation of the 18-bit DAC output in RNC. (b) Voltage dependency of ideal, nonlinear, and RNC-calibrated nonlinear resistor at 4 K. (c) Simulated DNL of 18-bit DAC w/ R nonlinearity at 4 K. (d) Simulated INL of 18-bit DAC w/ R nonlinearity at 4 K.

the triode region, M1 and M2 should have sufficient tunability of resistance to counteract the nonlinearity ($200 \sim 800 \Omega$) and mismatch of polysilicon resistors. A larger dimension of W/L leads to a steeper resistance transition, which needs a biasing DAC with higher resolution and suffers from increased sensitivity to noise. Simulation results in Fig. 10 show that by adjusting the gate-source voltages V_{gs} of M1 and M2 within the range of $0 \sim 0.7 \text{ V}$, the ON-resistance can be adjusted from 150Ω to $4 \text{ k}\Omega$ for M1, and from 120Ω to $2 \text{ k}\Omega$ for M2. In addition, the adjustment step of the ON-resistance R_{step} affects the effectiveness of RNC. The two input codes with the worst INL caused by R nonlinearity, shown in Fig. 9, are calibrated using different R_{step} . It is shown that the smaller R_{step} , the better the calibration effect on INL [Fig. 11(a)], with the smaller the adjustment step of 18-bit DAC output voltage V_{DAC_step} [Fig. 11(b)]. In order to calibrate the INL of the 18-bit DAC to be less than 1 LSB, R_{step} should be less than 20Ω , giving V_{DAC_step} of $<1 \text{ LSB}$ ($4.6 \mu\text{V}$). Subsequently, the auxiliary DACs need sufficient resolution to meet this requirement of R_{step} . Meanwhile, it shall have an intrinsic INL of $<1 \text{ LSB}$ even under the deteriorated nonlinearity and mismatch at cryogenic temperature, which needs no more calibration. Thus, the number of auxiliary DACs' bits should not be set too high so that INL becomes unacceptable, while also considering the limited storage space for calibration codes. Consequently, the resolution of auxiliary DACs is determined to be 10-bit. The simulated INL of the 10-bit DAC is 0.1 LSB [Fig. 11(c)]. Thus, to the first-order, the ON-resistance of M1 and M2 can be precisely tuned with a precision of $0.1\text{--}15 \Omega$ within $200 \sim 800 \Omega$, leaving sufficient margin for RNC calibration.

Besides a segmented R-2R ladder, RNC also adopts a segmentation scheme to reduce the storage space of calibration codes. First, as shown in Fig. 12(a), the full range of the DAC

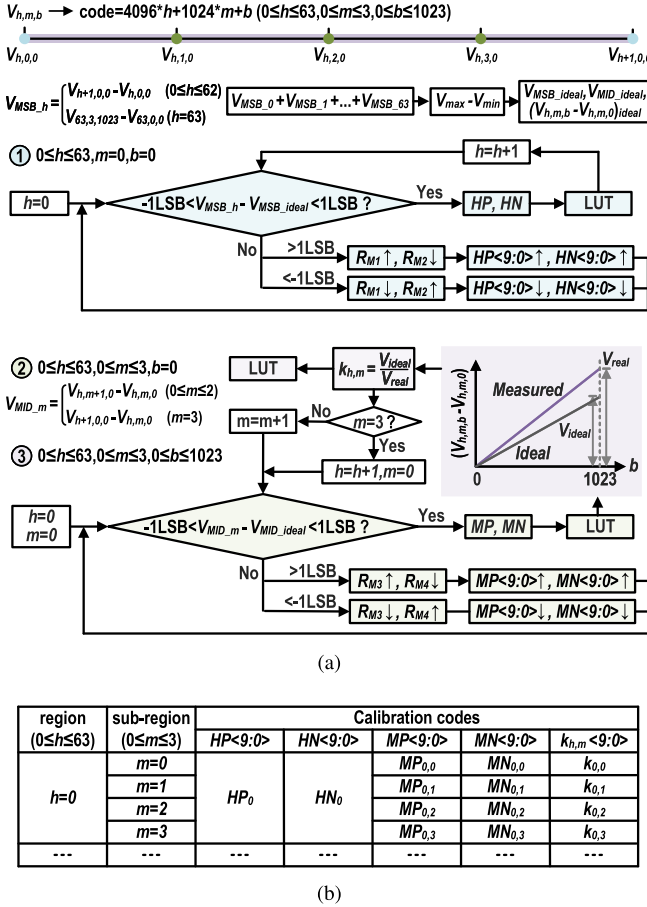


Fig. 13. (a) Calibration procedure of RNC. (b) Storage of RNC calibration codes.

output is divided into 64 equally spaced regions by $H < 62:0 >$. Next, the ON-resistance of M1 and M2 is fine-tuned in each of the 64 MSB regions by the two auxiliary 10-bit DACs to compensate for the resistor nonlinearity. This reduces the INL by a factor of approximately $64 \times$ [Fig. 12(b)]. In addition, when $H < 0 >$ is connected to "0," a higher compensation resistance of M2 is required in the lower output regions. Since the ON-resistance exhibits a strong nonlinear variation with the gate voltage, this leads to fluctuations of the calibrated nonlinearity. In contrast, the larger ON-resistance of M1 is demanded at higher output regions when $H < 0 >$ is connected to "1," also resulting in fluctuations. Subsequently, the MID thermal code $M < 2:0 >$ further divides each of the 64 regions into four sub-regions, where the re-compensation will further reduce the nonlinearity error.

For the RNC calibration at cryogenic temperatures, the 18-bit DAC operates in Mode 2: two-point modulation, where V_{DAC} is periodically toggled between voltages corresponding to two specific DAC input codes: V_{c1} and V_{c2} . The relative voltage of V_{c1} and V_{c2} is measured through lock-in detection, which suppresses low-frequency noise and interference from the refrigerator. However, since the lock-in amplifier can only measure relative values rather than absolute values, a fixed voltage reference should be established. The V_{DAC} at zero input (V_{min}) is regarded as the reference voltage, at which point V_{DAC} is equal to the ground voltage and not affected by R

nonlinearity. Then, the relative voltage between the full-scale output of DAC V_{max} and V_{min} is first determined, and subsequently, RNC is performed within this range. It is obtained by accumulating the measured voltage spans of 64 MSB regions, owing to the superior accuracy of lock-in detection with small-amplitude inputs. This voltage span $V_{MSB,h}$ refers to the relative output voltage between the starting input codes of adjacent regions [Fig. 13(a)]. As for the final region, its voltage span $V_{MSB,63}$ is defined as the relative voltage between its starting and final input codes, which is 1 LSB smaller than the others. This corresponds to the V_{max} being 1 LSB away from AVDD. When $V_{MSB,63}$ is defined as the relative voltage between the output voltage of the starting code and AVDD, the spans of all regions are theoretically identical. Once the overall output range of the DAC is determined, the theoretical relative voltage between two input codes can be established.

First, perform the RNC on the starting input code of each region $V_{h,0,0}$. With $V_{0,0,0}$ (i.e., V_{min}) as the reference, $V_{1,0,0}$ is calibrated by comparing the measured $V_{MSB,0}$ with the theoretical voltage span of the first region. If $V_{MSB,0}$ is larger, increase the ON-resistance of the switches connected to AVDD and decrease that of the switches connected to ground. Thus, the reverse adjustments apply for the lower $V_{MSB,0}$. When the DNL and INL are both within ± 1 LSB, the calibration codes can be confirmed. Then, the calibrated $V_{1,0,0}$ serves as the reference for calibrating $V_{2,0,0}$ of the next region, and this process repeats across all regions. Second, the starting input codes of four sub-regions in each region $V_{h,m,0}$ are calibrated in the same manner, with the calibrated $V_{h,0,0}$ as the initial reference. Third, remaining input codes in each sub-region are affected by the residual nonlinearity of $B < 9:0 >$. As $B < 9:0 >$ increases, the thermal code resistors connected to ground increase with V_{DAC} , in contrast to the resistors connected to AVDD that decrease. This results in the actual transfer characteristic of the DAC exhibiting a steeper slope than the ideal within the sub-region. The slope deviation can be quantified, yielding the gain factor $k_{h,m}$ for the slope calibration. This calibration is performed with the 3-bit fine calibration code $L < 2:0 >$, which is equivalent to the fractional part of the input code and can adjust V_{DAC} in steps of $1/8$, $1/4$, and $1/2$ LSB. Finally, all the calibration codes are stored in a lookup table (LUT) in the segmented manner, as illustrated in Fig. 13(b). Each region holds 2 10-bit MSB calibration codes, 8 10-bit MID calibration codes, and 4 10-bit gain factors for the binary segment $B < 9:0 >$, requiring a total storage capacity of 1.12 kB. The simulated DNL in Fig. 12(c) and INL in Fig. 12(d) indicate that RNC can effectively reduce the resistor nonlinearity at cryogenic temperature.

C. Ordered Element Matching

Although the polysilicon resistor array is carefully sized and layouted with surrounding dummies, the deteriorated device matching still limits the DAC resolution after cooling (Fig. 9). Instead of the conventional DEM, which introduces random noise, the static OEM is applied on the thermal code resistors [40]. It minimizes the random mismatch by rearranging the 63-bit MSB thermal code resistors into a 6-bit binary code array with enhanced linearity, significantly reducing chip area

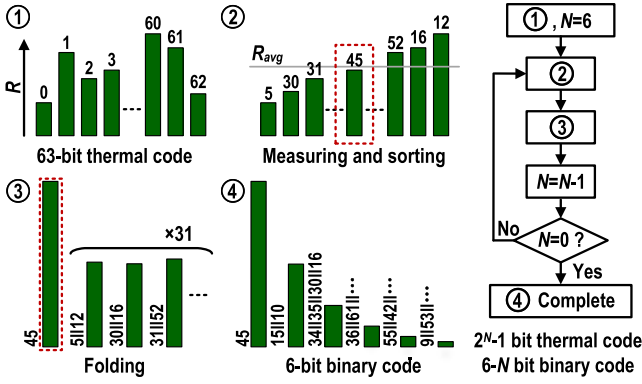


Fig. 14. Calibration procedure of OEM with successive folding.

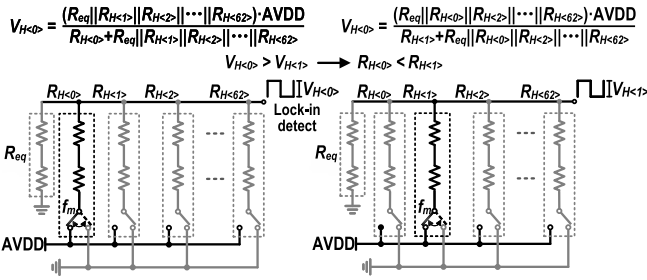


Fig. 15. Resistance measurement of OEM, where R_{eq} indicates the equivalent resistance when $L<2:0>$, $B<9:0>$, and $M<2:0>$ are all connected to the ground.

and digital circuit overhead. As shown in Fig. 14, rectangles indicate the 63 thermal code resistors with random mismatch errors in the MSB segment. In the OEM calibration scheme, first, the 63 thermal code resistors are labeled as No. 0–No. 62. Second, they are sorted according to their resistance. The resistor (for instance, No. 45), which is closest to the average resistance R_{avg} , is chosen as the LSB of a 6-bit binary code. Third, 31 pairs of resistors with a net resistance closer to $R_{avg}/2$ are selectively folded, which yields an array with a 1-bit binary code and a 31-bit thermal code with a reduced deviation. As this process repeats, the mismatch errors are consistently diminishing. Finally, the 63-bit thermal code $H<62:0>$ is turned into a 6-bit binary code with improved linearity. Similarly, the 3-bit MID thermal code $M<2:0>$ is turned into a 2-bit binary code.

Precise measurement of the relative resistance of thermal code resistors is critical for the OEM calibration. During the first round of OEM calibration, only the thermal code resistor under test $R_{H< i >}$ ($i = 0, \dots, 62$) is toggled between AVDD and ground with a modulation frequency of f_m , while all of the rest resistors are wired to ground, as shown in Fig. 15. All pMOS and nMOS switch pairs (including the one associated with the resistor under measurement) exhibit the minimum ON-resistance, where the adjustment of switch resistance in RNC is not activated. Then, the DAC output voltage V_{DAC} is measured through lock-in detection at f_m . Since V_{DAC} is determined by the division ratio of thermal code resistors, the higher the V_{DAC} , the smaller the resistor under test $R_{H< i >}$. Then, by measuring all of the resistors sequentially, the first round of sorting and folding can be performed (Fig. 14).

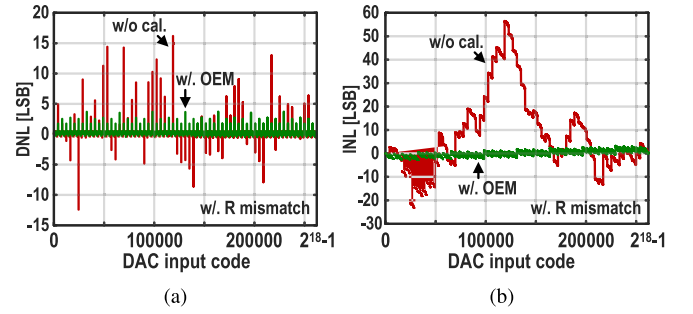


Fig. 16. Simulated (a) DNL and (b) INL of 18-bit DAC at 4 K with R mismatch only.

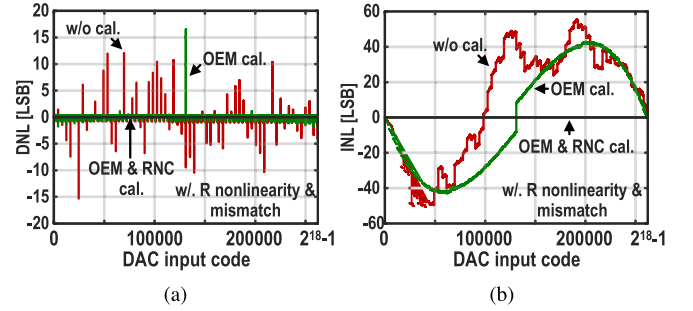


Fig. 17. Simulated (a) DNL and (b) INL of 18-bit DAC at 4 K with R nonlinearity and mismatch.

Next, after measuring the selected 31 pairs of resistors in the similar manner, the second round of sorting and folding can be performed. By repeating this process, the binary combination of thermal code resistors can be obtained. Fig. 16 presents the simulated calibration results of OEM on resistor mismatch at 4 K. Both the DNL and INL are significantly reduced. The mapping relationship of a 63-bit thermal code and a 6-bit binary code is similarly stored in an LUT.

Finally, Fig. 17 presents the simulated DNL and INL of an 18-bit DAC with R nonlinearity and mismatch at 4 K after employing OEM and RNC, which can be trimmed to be less than 1 LSB. This validates the feasibility of the two calibration methods at cryogenic temperatures. In addition, compared to other popular calibration techniques, OEM and RNC present the advantages of introducing no additional noise and consuming ultra-LP, critical for the control of quantum devices. The DEM averages the mismatch through dynamically changing the positions of elements, but would introduce the random voltage noise [43]. The 20-bit DAC in [24] employs an auxiliary IDAC and force and sense technique assisted by operational amplifiers, requiring higher DC power and circuit complexity. The on-chip self-calibration of DACs also requires the error detection circuits with extra DC power, whose accuracy also hardly supports an 18-bit DAC [44], [45].

V. MEASUREMENT RESULTS

This chip is fabricated using the standard 65-nm CMOS LP process with a chip area of 2×2 mm. The die photograph and chip packaging are shown in Fig. 18. It is controlled by the serial-to-parallel interface (SPI) with a highest speed of 1 Mbit/s. The digital clock of this chip can be tuned within

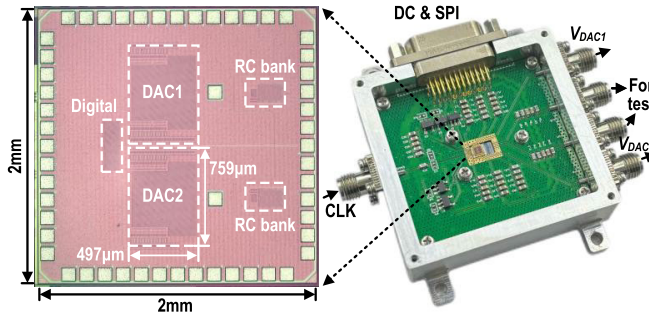


Fig. 18. Microphoto of the 65-nm Cryo-CMOS dual 18-bit DAC manipulator chip and packaging.

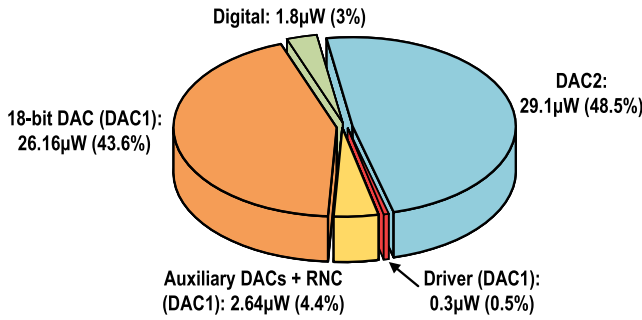


Fig. 19. Power consumption breakdown. The total DC power is 60 μ W.

$f_{clk} \leq 60$ MHz. With a supply voltage of 1.2 V, it consumes a DC power P_{DC} of 60 μ W under Mode 1: DC biasing mode, as broken down in Fig. 19. P_{DC} raises to 300 μ W under Mode 3: three-point modulation with the highest modulation frequency of 20 MHz at $f_{clk} = 60$ MHz.

A. Electrical Performance Characterization

The Cryo-CMOS chip is mounted on a 4-K pulse tube refrigerator for the DNL and INL measurement of the DAC. An 81/2 digital multimeter is a common approach for the measurement of high precision DAC [Fig. 20(a)]. However, the vibration, noise, and interference of the refrigerator deteriorate the low-frequency noise spectrum of the ground. In practice, an integrated ground noise of 15 μ V from 1 Hz to 50 kHz is measured, which makes the calibration unrepeatable (1 LSB = 4.6 μ V). Therefore, Mode 2: two-point modulation is adopted to suppress the ground noise. After completing the OEM and RNC segmented calibration, perform the segmented sequential measurement on the 64 MSB regions with the calibration codes from the LUT. Within each of the regions, one input code is fixed as the starting input code while the other is sequentially stepping up until the starting code of the next region, ensuring no breakpoints between adjacent regions. These two codes are toggled periodically with a modulation frequency of $f_m = 10$ kHz. The DAC output is detected by a lock-in amplifier [Fig. 20(b)]. Afterward, the DNL between any two adjacent input codes of the DAC can be obtained, and the INL is then derived by accumulating the DNL. For a fair comparison, the measurement results of INL using an 81/2 digital multimeter and lock-in detection with Mode 2 at 300 K, where no ground noise is present, are shown in

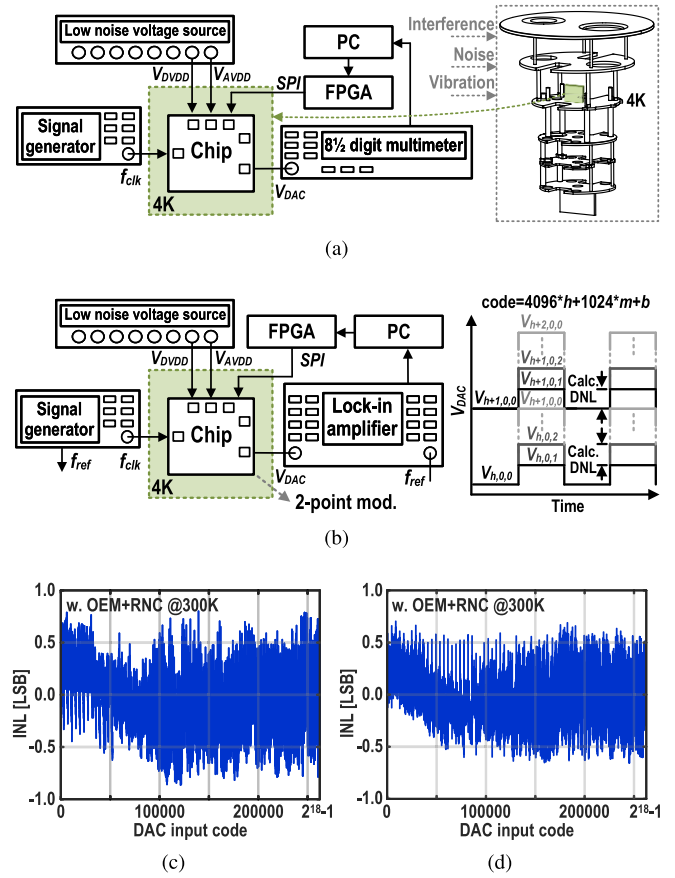


Fig. 20. Measurement setup of DAC DNL and INL. (a) w/ 81/2 digital multimeter. (b) w/ lock-in amplifier. Measured INL of DAC w/ calibration at 300 K (c) via 81/2 digital multimeter and (d) via lock-in amplifier.

Fig. 20(c) and (d). The results coincide with each other very well, which validates the effectiveness of Mode 2.

First, without calibration, the measurement at 4 K indicates a DNL of ± 5 –10 LSB [Fig. 21(a)] and an INL of ± 25 –45 LSB [Fig. 21(b)] for the 18-bit DAC. After the OEM is applied, the measured DNL and INL are reduced to ± 1 –5 LSB and ± 5 –30 LSB, respectively. Next, after applying the RNC, the final DNL and INL of DAC are reduced to ± 0.8 LSB. In addition, using the same calibration codes at 4 K, variations of DNL and INL with the environment temperature T_{env} are also measured from 4 to 10 K. Fig. 21(c) and (d) indicates a degraded DNL of ± 6 –2 LSB and an INL of ± 5 –8 LSB at 10 K, which can be explained by the variation of resistor nonlinearity. Fortunately, considering the ± 0.1 -K temperature control capability of the dilution refrigerator, it will not affect the resolution of DAC. Please also note that the 4-K measurements in Fig. 21(c) and (d) are newly obtained data, coinciding with the initial calibration a couple of days ago in Fig. 21(a) and (b) very well. Then, multiple DACs are measured to verify the reliability of the calibration. The results in Fig. 21(e) and (f) demonstrates that despite mismatch differences among multiple DACs, the DNL and INL can be calibrated to ± 0.8 LSB. Furthermore, the resistance distribution of thermal code resistors can be obtained. During the OEM calibration, one of the 63 MSB thermal code resistors is sequentially connected to AVDD,

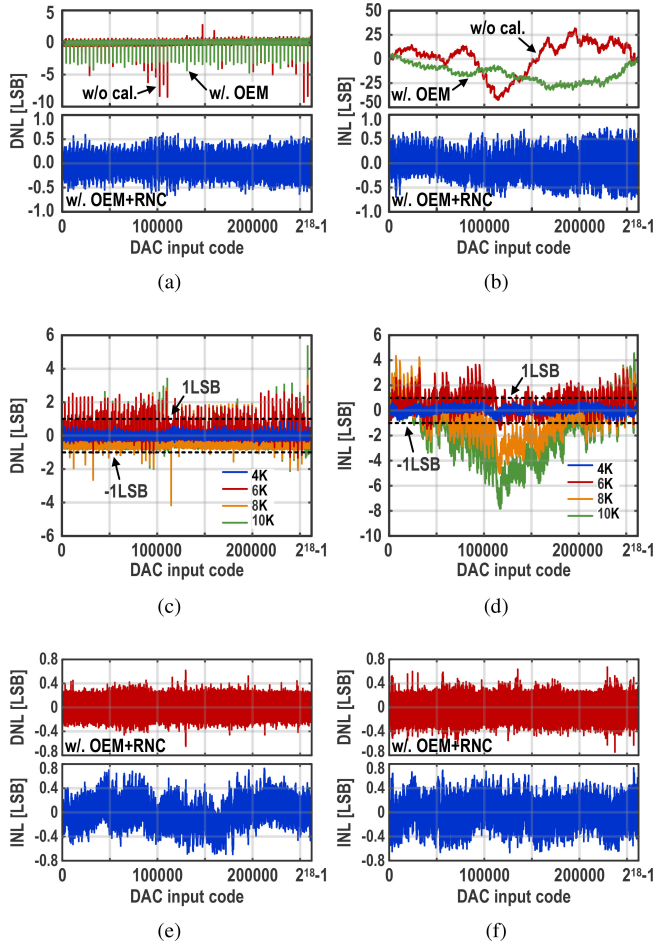


Fig. 21. Measured linearity of three DACs. (a) DNL of the first DAC at 4 K. (b) INL of the first DAC at 4 K. (c) DNL and (d) INL variation of the first DAC over temp. $T_{env} = 4 \sim 10$ K. (e) DNL and (f) INL of the second DAC at 4 K. (g) DNL and (h) INL of the third DAC at 4 K.

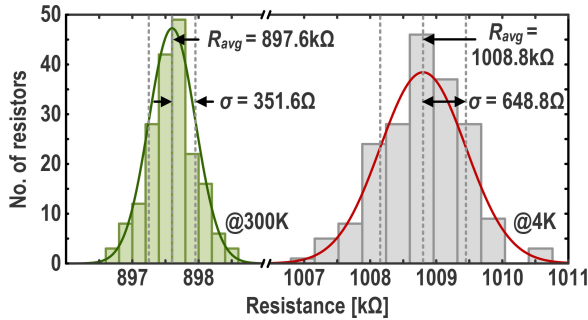


Fig. 22. Resistance distribution of MSB thermal code resistors in three DACs at 300 and 4 K.

while the rest are connected to ground. Its resistance value can be determined based on the output voltage (Fig. 15) and the measured average resistance [Fig. 7(d)]. The resistance distribution of 189 thermal code resistors (8R) in three DACs at 300 and 4 K is shown in Fig. 22. The standard deviation (σ) of the resistor mismatch at 4 K is 648.8 Ω , which is approximately $2\times$ worse than that of 351.6 Ω at 300 K. Since each thermal code resistor consists of eight identical resistors R , its standard deviation is the result of averaging the random deviations of them. Thus, the standard deviation of R should be $\sqrt{8}\sigma$, which is 994.5 Ω at 300 K and 1.8 k Ω at

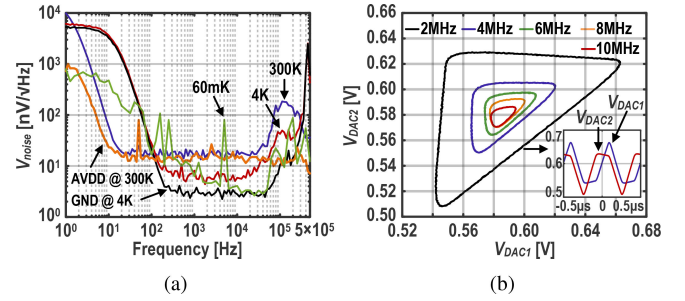


Fig. 23. Measured noise spectrum and three-point modulation of 18-bit R-2R DAC. (a) Output noise spectrum density. (b) Three-point modulation at 4 K.

4 K. While this standard deviation is affected by the sample quantity and the measurement system, it clearly indicates that the resistor mismatch deteriorates by approximately $2\times$ at 4 K, demonstrating that the assumption of a $3\times$ deterioration in resistor mismatch simulated in Fig. 8 is reasonable.

Second, the output noise spectrum of the 18-bit DAC, the internal ground of the refrigerator, and an external low-noise voltage source are measured using a lock-in amplifier. To eliminate the noise impact of AVDD, the DAC operates at the zero input state. Fig. 23(a) shows that the noise spectrum density of DAC at 300 K is 19.9 nV/Hz^{0.5} with a baseband frequency of 10 kHz. The ground of the refrigerator exhibits significant low-frequency noise, which dominates the output noise of the DAC at 4 K, and is the major reason that Mode 2: two-point modulation is applied. After deembedding the ground noise of the refrigerator, the noise spectrum density of DAC at 4 K is 5.9 nV/Hz^{0.5} and the integrated noise from 1 Hz to 50 kHz is 2.6 μ V. When the DAC biases the gate of silicon SET or QD, the voltage noise is limited by both the thermal noise kT/C and the low-frequency flicker noise, where C is the load capacitor. The chosen noise integration bandwidth of 50 kHz encompasses the thermal and flicker noise, excluding the significant ground noise of the refrigerator at higher frequencies. Measured at 60 mK, the voltage noise density achieves 4.1 nV/Hz^{0.5} and the integrated noise is 3.8 μ V, which are still limited by the residual ground noise of the dilution refrigerator.

Third, the three-point modulation at 4 K is measured in Fig. 23(b), which is originally designed for the single-electron transportation control of a SEP device. V_{DAC1} and V_{DAC2} follow a triangular-shaped path periodically in the voltage space. The modulation speed is 2–10 MHz. The shrinking DAC amplitude at higher frequency is due to the ~ 200 -pF load capacitance of the 2-m long, 50 Ω interconnection cable from 4 to 300 K, which leads to a time constant of $\tau = 3.2 \mu$ s. In future applications, it can be improved by connecting to quantum devices with a <1 -pF load capacitor.

B. Experiment With Si SET at 60 mK

The Cryo-CMOS chip is co-integrated with the Si SET at 60 mK in the dilution refrigerator, as shown in Fig. 24(a). The distance between them is approximately 45 mm, connected via coaxial cables. Fig. 24(b) illustrates the zoom-in-view detail of the chip and the STM photograph of SET.

TABLE I
PERFORMANCE COMPARISON WITH THE OTHER STATE-OF-THE-ARTS

Parameters	This work			VLSI 2022 [12]		SSCL 2020 [8]		IEICE 2022 [19]	PRA 2019 [26]	BCICTS 2023 [27]	ISSCC 2013 [24]
Quantum device	Si SET			N/A		Si SET		Si QD	Trapped ions	N/A	N/A
Temp. T_{env} (K)	300	4	0.06	300	4.2	6	0.44	8	77	300	300
DAC architecture	R-2R Ladder			CDAC w/. Op-amp		CDAC w/o Op-amp		CDAC w/. Op-amp	R-2R + Op-amp	R-2R + Op-amp	R-2R Ladder
DAC calibration	RNC + OEM			N/A		N/A		Capacitor mismatch cal.	Offset cal.	N/A	IDAC + force & sence
Resolution (Bit)	18			15		13		11	12	12	20
DNL (LSB)	0.8			0.6	0.8	2.5		2	N/A	N/A	0.30
INL (LSB)	0.8			12.6	36.5	N/A		2	N/A	N/A	0.35
Precision V_{LSB} (μ V)	4.58			68.3	57.1	122		1221	3907	440	19
Int. $V_{noise}^{\dagger}$ (μ V $_{rms}$)	8.6	2.6	3.8	192	188	N/A		N/A	N/A	32.1	1.7
V_{noise} (nV/Hz $^{0.5}$) ††	19.9	5.9	4.1	N/A		N/A		N/A	10000	80	7.5
Output range (V)	1.2			3		1		2.5	16	1.8	20
No. of channels	2			1		8 †††		1	16	1	1
Modulation	2/3-point mod.			N/A		N/A		N/A	N/A	N/A	N/A
P_{DC} (μ W)	61.6	52.1	59.9	138	157	21.1	20	5.8	31250	1100	84000
Area/ch. (mm 2)	0.377			0.076		0.14		0.008	0.035	0.84	8.54
Technology	65 nm CMOS			22 nm FinFET		65 nm CMOS		40 nm CMOS	180 nm CMOS	130 nm CMOS	0.6 μ m BiCMOS

† Integrated from 1 Hz to 50 kHz. †† @ 10 kHz. ††† Single DAC demultiplexed to 8 channels.

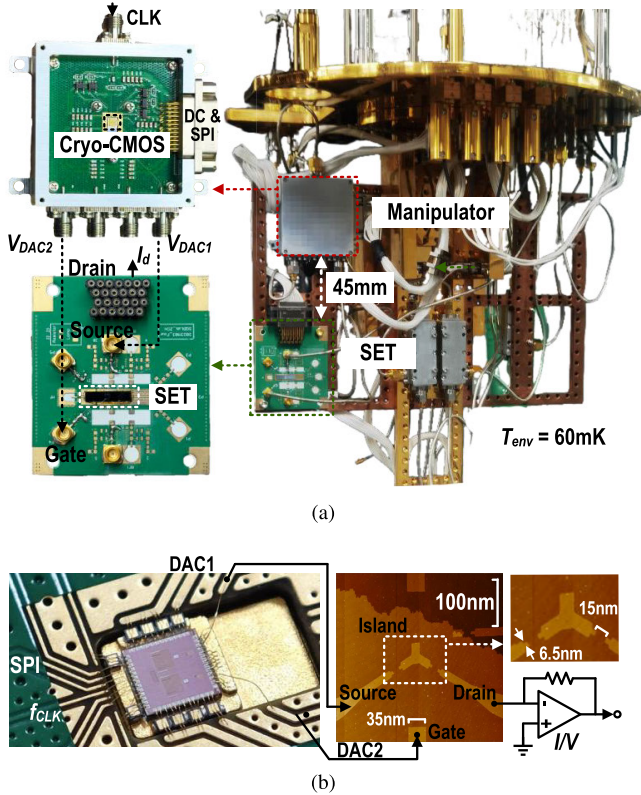


Fig. 24. Co-integration of Cryo-CMOS chip and Si SET at 60 mK in the dilution refrigerator. (a) Measurement setup. (b) Zoom-in-view detail of Cryo-CMOS chip and the STM photograph of SET.

The SET consists of four core components: two source/drain electrodes, a QD island, and a gate electrode. The width of the source/drain electrodes is 6.5 nm, with a gap of about 15 nm from the island, and the width of the gate electrode is 45 nm. The dimension of the island is several tens of nanometers,

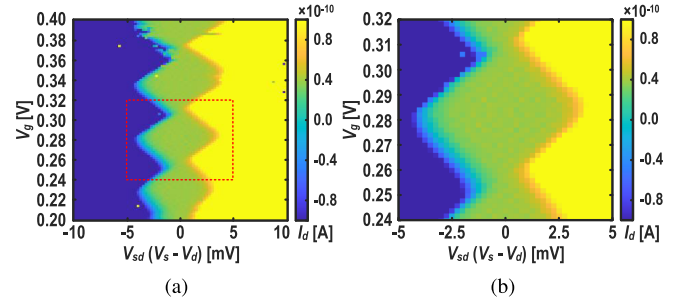


Fig. 25. (a) Measured coulomb diamond of SET at 60 mK. (b) Zoom-in-view detail of the coulomb diamond at 60 mK.

which is smaller than the de Broglie wavelength of electrons, exhibiting a significant quantum confinement effect. The island is capacitively coupled to the electrodes. Due to the presence of Coulomb repulsion, electrons can only tunnel under special conditions. By adjusting the gate voltage, the electrochemical potential of the QD can be varied continuously. When the energy level of the QD aligns with or lies within the Fermi levels of the source and drain, the Coulomb blockade is lifted, allowing the tunneling of electrons and generating a current peak. With the scanning of gate voltage, periodic current peaks appear, corresponding to a single electron being precisely added into the QD, thereby achieving control over the transport of electrons.

When manipulating SET with the Cryo-CMOS chip at 60 mK, the two DACs operate in Mode 1: DC biasing, controlling the source and gate of the SET, respectively. The drain of the SET is connected to the TIA (gain: 10^9 V/A) at 300 K for the current-to-voltage conversion, enabling drain current readout. DAC1 sets the source voltage of SET ($-10 \sim +10$ mV) to establish the energy gap between source and drain, while DAC2 scans the gate voltage, regulating the electrochemical

potential of the QD. The 2-D scanning maps the boundaries of the Coulomb blockade regions, which manifests as a clear Coulomb diamond as shown in Fig. 25. This validates the effectiveness of the Cryo-CMOS chip, which could control the transport of electrons in SET. Furthermore, based on the full-width at half maximum (FWHM) of the Coulomb peak measured under the ultra-low source-drain bias, the electron temperature of SET is approximately 900 mK, while the more precise value requires further temperature-dependent experiments.

VI. CONCLUSION

This article presents a Cryo-CMOS dual 18-bit R-2R DAC manipulator co-integrated with the Si SET at 60 mK. Table I summarizes the performance of this work with the other state-of-the-art. After the proposed RNC and OEM calibration of the resistor nonideality, it achieves an unprecedented 18-bit resolution with a minimum step of $4.6 \mu\text{V}$. It also achieves so far the lowest integrated noise of $3.8 \mu\text{V}$ and noise spectrum density of $4.1 \text{ nV}/\text{Hz}^{0.5}$ at 60 mK. It only burns a DC power of $60 \mu\text{W}$. The Cryo-CMOS chip has been co-integrated with the SET at 60 mK and achieved precise control of it, obtaining a clear Coulomb diamond that validated the effectiveness of the chip. This work is promising to provide the precise gate biasing for large-scale silicon QDs in a multiplexing manner (Fig. 1) to control their electrochemical potentials and coupling, and allows the co-integration at a temperature level of millikelvin.

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REFERENCES

- [1] J.-S. Park et al., "13.1 A fully integrated cryo-CMOS SoC for qubit control in quantum computers capable of state manipulation, readout and high-speed gate pulsing of spin qubits in Intel 22 nm FFL FinFET technology," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, vol. 64, Feb. 2021, pp. 208–210.
- [2] A. G. Fowler, M. Mariantoni, J. M. Martinis, and A. N. Cleland, "Surface codes: Towards practical large-scale quantum computation," *Phys. Rev. A, Gen. Phys.*, vol. 86, no. 3, Sep. 2012, Art. no. 032324.
- [3] J. Anders et al., "CMOS integrated circuits for the quantum information sciences," *IEEE Trans. Quantum Eng.*, vol. 4, pp. 1–30, 2023.
- [4] L. M. K. Vandersypen et al., "Interfacing spin qubits in quantum dots and donors—Hot, dense, and coherent," *NPJ Quantum Inf.*, vol. 3, no. 1, Sep. 2017, Art. no. 34.
- [5] S. Neyens et al., "Probing single electrons across 300-mm spin qubit wafers," *Nature*, vol. 629, no. 8010, pp. 80–85, May 2024.
- [6] M. Fuechsle et al., "A single-atom transistor," *Nature Nanotechnol.*, vol. 7, no. 4, pp. 242–246, 2012.
- [7] T. Berer et al., "Single-electron transistor in strained Si/SiGe heterostructures," *Phys. E, Low-Dimensional Syst. Nanostruct.*, vol. 34, nos. 1–2, pp. 456–459, Aug. 2006.
- [8] P. Vliex et al., "Bias voltage DAC operating at cryogenic temperatures for solid-state qubit applications," *IEEE Solid-State Circuits Lett.*, vol. 3, pp. 218–221, 2020.
- [9] M. D. Reed et al., "Reduced sensitivity to charge noise in semiconductor spin qubits via symmetric operation," *Phys. Rev. Lett.*, vol. 116, no. 11, Mar. 2016, Art. no. 110402.
- [10] T. Futaya et al., "Cryogenic flip-chip interconnection for silicon qubit devices," *Jpn. J. Appl. Phys.*, vol. 63, no. 3, Mar. 2024, Art. no. 03SP64.
- [11] S. K. Barte et al., "Spin-qubit control with a milli-kelvin CMOS chip," *Nature*, vol. 643, no. 8071, pp. 382–387, Jul. 2025.
- [12] L. Enthoven, J. van Staveren, J. Gong, M. Babaie, and F. Sebastiano, "A 3 V 15b 157 μW cryo-CMOS DAC for multiplexed spin-qubit biasing," in *Proc. IEEE Symp. VLSI Technol. Circuits (VLSI Technol. Circuits)*, Jun. 2022, pp. 228–229.
- [13] L. Schreckenberger et al., "SiGe qubit biasing with a cryogenic CMOS DAC at mK temperature," in *Proc. IEEE 49th Eur. Solid State Circuits Conf. (ESSCIRC)*, Sep. 2023, pp. 161–164.
- [14] S. Pauka et al., "A cryogenic CMOS chip for generating control signals for multiple qubits," *Nature Electron.*, vol. 4, no. 1, pp. 64–70, 2021.
- [15] A. Caglar et al., "A scalable mK DC demultiplexer with extremely low OFF-leakage CMOS switches for biasing of spin qubits," *IEEE Solid-State Circuits Lett.*, vol. 8, pp. 221–224, 2025.
- [16] J. van Staveren et al., "Cryo-CMOS bias-voltage generation and demultiplexing at mK temperatures for large-scale arrays of quantum devices," *IEEE Trans. Quantum Eng.*, vol. 6, pp. 1–18, 2025.
- [17] S. Park, S. Song, K. Kim, H. Lee, M. Lee, and J.-Y. Sim, "A cryo-CMOS 64-channel bias generator IC for surface ion trap," in *Proc. IEEE Eur. Solid-State Electron. Res. Conf. (ESSERC)*, Sep. 2024, pp. 488–491.
- [18] R. Otten et al., "Qubit bias using a CMOS DAC at mK temperatures," in *Proc. 29th IEEE Int. Conf. Electron., Circuits Syst. (ICECS)*, Oct. 2022, pp. 1–4.
- [19] T. Miki, R. Takahashi, and M. Nagata, "An 11-bit 0.008 mm² charge-redistribution digital-to-analog converter operating at cryogenic temperature for large-scale qubit arrays," *IEICE Electron. Exp.*, vol. 19, no. 8, 2022, Art. no. 20220099.
- [20] S. Das and J. C. Bardin, "Characterization of shot noise suppression in nanometer MOSFETs," in *IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2021, pp. 892–895.
- [21] R. Asanovski et al., "Understanding the excess 1/f noise in MOSFETs at cryogenic temperatures," *IEEE Trans. Electron Devices*, vol. 70, no. 4, pp. 2135–2141, Apr. 2023.
- [22] M. E. P. V. Zurita et al., "Cryogenic current steering DAC with mitigated variability," *IEEE Solid-State Circuits Lett.*, vol. 3, pp. 254–257, 2020.
- [23] J. Park et al., "A fully integrated cryo-CMOS SoC for state manipulation, readout, and high-speed gate pulsing of spin qubits," *IEEE J. Solid-State Circuits*, vol. 56, no. 11, pp. 3289–3306, Nov. 2021.
- [24] R. C. McLachlan, A. Gillespie, M. C. W. Coln, D. Chisholm, and D. T. Lee, "A 20b clockless DAC with sub-ppm-linearity 7.5nV/vHz-noise and 0.05ppm/°C-stability," in *Proc. IEEE Int. Solid-State Circuits Conf. Dig. Tech. Papers*, Feb. 2013, pp. 278–279.
- [25] R. C. McLachlan, A. Gillespie, M. C. W. Coln, D. Chisholm, and D. T. Lee, "A 20b clockless DAC with sub-ppm INL, 7.5 nV/vHz noise and 0.05 ppm/°C stability," *IEEE J. Solid-State Circuits*, vol. 48, no. 12, pp. 3028–3037, Dec. 2013.
- [26] J. Stuart et al., "Chip-integrated voltage sources for control of trapped ions," *Phys. Rev. Appl.*, vol. 11, no. 2, Feb. 2019, Art. no. 024010.
- [27] A. Meyer, P. Toth, A. Engelhardt, J. Repp, M. Brand, and V. Issakov, "A 12 bit R-2R digital-to-analog converter for shuttling operation in a trapped-ion quantum computer," in *Proc. IEEE BiCMOS Compound Semiconductor Integr. Circuits Technol. Symp. (BCICTS)*, Oct. 2023, pp. 28–31.
- [28] E. Charbon et al., "15.5 cryo-CMOS circuits and systems for scalable quantum computing," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2017, pp. 264–265.
- [29] H.-M. Chuang et al., "A comprehensive study of polysilicon resistors for CMOS ULSI applications," *Superlattices Microstructures*, vol. 33, no. 4, pp. 193–208, Apr. 2003.
- [30] M. Nakanishi, J. Kinoshita, Y. Murayama, and T. Endo, "Current dependence measurement of resistance using a cryogenic constant current source with linear output," in *Conf. Precis. Electromagn. Meas. Conf. Dig.*, 2000, pp. 303–304.
- [31] A. Grill et al., "Temperature dependent mismatch and variability in a cryo-CMOS array with 30k transistors," in *Proc. IEEE Int. Rel. Phys. Symp. (IRPS)*, Mar. 2022, pp. 10A.1-1–10A.1-6.
- [32] Y. Li, Y. Zhang, H. Lin, and C. Wang, "13.4: Xiling: Cryo-CMOS 18-bit dual-dac manipulator with 4.6 μV precision and 4.1nV/Hz^{0.5} noise co-integrated with the single electron transistor at 60mK," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2025, pp. 242–244.
- [33] J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, "Single-shot read-out of an individual electron spin in a quantum dot," *Nature*, vol. 430, no. 6998, pp. 431–435, Jul. 2004.

- [34] R. Hanson, L. P. Kouwenhoven, J. R. Petta, S. Tarucha, and L. M. K. Vandersypen, "Spins in few-electron quantum dots," *Rev. Mod. Phys.*, vol. 79, no. 4, pp. 1217–1265, Oct. 2007.
- [35] A. Morello et al., "Single-shot readout of an electron spin in silicon," *Nature*, vol. 467, no. 7316, pp. 687–691, Sep. 2010.
- [36] Y. V. Nazarov and Y. M. Blanter, *Quantum Transport: Introduction to Nanoscience*. Cambridge, U.K.: Cambridge Univ. Press, 2009.
- [37] G. Yamahata, S. P. Giblin, M. Kataoka, T. Karasawa, and A. Fujiwara, "High-accuracy current generation in the nanoampere regime from a silicon single-trap electron pump," *Sci. Rep.*, vol. 7, no. 1, p. 45137, Mar. 2017.
- [38] B. Roche et al., "A two-atom electron pump," *Nature Commun.*, vol. 4, no. 1, Mar. 2013, Art. no. 1581.
- [39] K. L. Chan, N. Rakuljic, and I. Galton, "Segmented dynamic element matching for high-resolution digital-to-analog conversion," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 55, no. 11, pp. 3383–3392, Dec. 2008.
- [40] Y. Li and D. Chen, "Low-cost, high-precision DAC design based on ordered element matching," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 66, no. 2, pp. 502–512, Feb. 2019.
- [41] F. J. De la Hidalga, M. J. Deen, and E. A. Gutierrez, "Theoretical and experimental characterization of self-heating in silicon integrated devices operating at low temperatures," *IEEE Trans. Electron Devices*, vol. 47, no. 5, pp. 1098–1106, May 2000.
- [42] J. Marqués-García, J. Pérez-Bailón, S. Celma, and C. Sánchez-Azqueta, "Characterization of 65-nm CMOS integrated resistors in the cryogenic regime," *IEEE Trans. Instrum. Meas.*, vol. 73, pp. 1–3, 2024.
- [43] D.-H. Lee, T.-H. Kuo, and K.-L. Wen, "Low-cost 14-bit current-steering DAC with a randomized thermometer-coding method," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 56, no. 2, pp. 137–141, Feb. 2009.
- [44] G. A. Miller, M. C. W. Coln, L. A. Singer, and P. R. Oaklander, "A true 16 b self-calibrating BiCMOS DAC," in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 1993, pp. 58–59.
- [45] Y. Cong and R. L. Geiger, "A 1.5-V 14-bit 100-MS/s self-calibrated DAC," *IEEE J. Solid-State Circuits*, vol. 38, no. 12, pp. 2051–2060, Dec. 2003.



Yingjie Li received the B.E. degree in electronic science and technology from the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2024, where she is currently pursuing the M.S. degree.

She has been developing the cryogenic CMOS integrated circuits for quantum computers since 2022, at the Integrated Physics Group (IPG), School of Electronic Science and Engineering, UESTC. Her research interests include cryo-CMOS mixed-signal circuits and systems for the gate control of large-scale Qubit arrays.



Yifei Zhang received the B.E. degree in micro-electronics science and engineering and the M.Sc. degree in quantum science and engineering from the Southern University of Science and Technology (SUSTech), Shenzhen, China in 2022 and 2025, respectively.

He has been researching the heterogeneous integration scheme of cryogenic CMOS and quantum dot device at the Silicon Quantum Dot Lab, Shenzhen International Quantum Academy, SUSTech. His research interests include cryogenic CMOS design and single-electron spin qubits.



Haichuan Lin received the B.S. degree from the University of Science and Technology of China (USTC), Hefei, China, in 2010, and the Ph.D. degree from the University of Chinese Academy of Sciences (UCAS), Beijing, China, in 2014.

He is currently the Chief Engineer of Chengdu Data Automation System Technologies, Chengdu, China. He has been developing multiple room-temperature, multi-channel, high precision instruments for quantum manipulation and sensing. His research interests include but not limited to large-scale electronic systems, multi-channels timing synchronization, and high precision RF/analog measurement.



Cheng Wang (Senior Member, IEEE) received the B.E. degree from Tsinghua University, Beijing, China, in 2008, the M.S. degree from China Academy of Engineering Physics (CAEP), Mianyang, China, in 2011, and the Ph.D. degree from Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, in 2020.

He was an Assistant Research Fellow at the Institute of Electronic Engineering, CAEP, from 2011 to 2015. He was a Research Scientist at Analog Devices, Inc. (ADI), Boston, MA, USA, in 2020.

He joined the University of Electronic Science and Technology of China (UESTC), Chengdu, China, in 2021, and is currently a Professor at the School of Electronic Science and Engineering, UESTC. His current research interests include the cryogenic CMOS integrated circuits/systems for quantum computers, molecular clock&spectroscopy, and silicon-based millimeter/terahertz phased array.

Dr. Wang received the 2018 Chinese Government Award for Outstanding Self-Financed Student Abroad. He received the MIT Microsystem Technology Laboratory (MTL) 2019 Fall Doctoral Dissertation Seminar Award (1 of 2 students per year in MTL). In 2020, he was granted the IEEE Solid-State Circuit Society (SSCS) Predoctoral Achievement Award. In 2022, he was recognized as "35 innovators under 35" (China) by the MIT technology review.